

The Lookout and the Captain: Conditioning a Large Language Model with a Small Vision Model over a Latent Bus

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2026-08-26

Abstract

A robot’s behavioural context should not have to live in prompt text, which is overridable and makes a model act on words rather than on what it sees, nor require running the largest model on every frame. We study a third channel. A small vision model (a person detector, a few million parameters, a few milliseconds) watches the camera and emits a *mode* — social, cautious, exploring — onto a latent bus: a packet generated by a small hypernetwork from the mode’s text and injected into the residual stream of a frozen vision-language model, which still sees the frame itself. Across three frozen hosts (LFM2.5-VL 450M and 3B, Qwen3.8-27B) and eleven registered cells with three door seeds each, five things hold. The bus carries modes at every scale, and at 27B a mode on the bus conditions behaviour more strongly than the same instruction in the prompt (leans 0.64/0.87/0.47 against text 0.53/0.48/0.13) while staying grounded on empty frames where the prompt does not. Swapping the oracle label for an off-the-shelf detector is behaviourally invisible: fidelity 0.00 at every seed of both LFM hosts and $-0.03/-0.05$ at 27B, bounded exactly by miss rate times miss cost across three lookouts. Against a direct contrary order in the prompt the mode owns most of the contested behaviour at 27B (0.84–1.48 ownership at either dose), and door training strength — not injection dose — is the lever that buys robustness for free (3B ownership $0.00 \rightarrow 1.47$ at $2\times$ steps, grounding unchanged; $1.5\times$ dose buys less and loses the frame). Grounding lives in the host’s eyes, not in the bus: lying detector boxes move a sighted host at most 6%, and a blinded host is unreachable at operating dose except through the one command-strength door, which then captures it completely — so doors stay below command strength for hosts that can lose their eyes. And the lookout can gate the captain’s compute: waking the large model only on a verdict change saves 83.5% of decodes at 96–99% agreement with every-frame decoding, one frame of lag. Carried into a rendered world with nothing retrained, the stack’s sim-to-real bridge decomposes: the lookout needs a mesh-level human (capsule primitives read as sports equipment, 0.00; one photoscan, 0.995), the captain’s perception needs nothing, and the doors need mixed-domain training at twice the steps — a recipe that dominates the original at the 3B and comes with a scale on it, trading at the 450M and curing one seed while damaging another at the 27B. Every cell was registered with its readings and bands before it ran; the registration protocol is offered as a methods contribution alongside the architecture.

1 Introduction

A robot that runs a language model has three places to put its behavioural context. It can write the context into the prompt — “be cautious, people are around” — which is cheap and is also fragile:

prompt text is overridable by whatever else enters the context, and §3 measures the specific way it fails, by making the model act on an instruction rather than on what it sees. It can run its largest model on every frame and let the model discover the context for itself, which is the most reliable arrangement and the most expensive, and §7 measures how much of that expense is wasted. Or it can give the context a channel of its own, separate from the prompt and cheaper than the frame. This paper is about the third channel and what it can and cannot carry.

The architecture has two parts and a seam. A small vision model — a person detector of a few million parameters, running in a few milliseconds — watches the camera and emits a *mode*: not words, not facts, but a behavioural stance, “social” or “cautious” or “exploring”. The mode travels over a latent bus: a packet, generated by a small hypernetwork from the mode’s text, injected into the residual stream of a frozen large model at a few depths. The large model sees the frame itself and answers the robot’s question, what to do next. The small model is the lookout on the mast; the large model is the captain with the chart and the eyes; the bus is the one channel between them, and the whole question of the paper is what should cross it.

The answer, across three hosts from 450M to 27B parameters and eleven registered experimental cells, comes in five claims. The bus carries modes at every scale, and at 27B a mode on the bus conditions behaviour more strongly than the same instruction as prompt text (§3). The switch from an oracle label to an off-the-shelf detector is behaviourally invisible, with the residual gap bounded provably by miss rate times miss cost, verified across three lookouts and three hosts (§4). Against a direct contrary order in the prompt the mode owns most of the contested behaviour, and the lever that buys more robustness for free is door training strength, not injection dose — dose buys the same thing and spends the model’s grip on the frame (§5). Grounding lives in the captain’s eyes and not in the bus: a false alarm from the lookout barely moves a sighted host, lying boxes are ignored, and a blinded host is unreachable through the bus except by the one door strong enough to act as a command channel, which then captures it completely — so the deployment rule is doors below command strength for any host that can lose its eyes (§6). And the lookout can gate the captain’s compute: waking the large model only on a verdict change saves 83.5% of its decodes at 96–99% agreement with every-frame decoding, one frame of lag, while an understanding-free pixel gate saves nothing, exactly as predicted (§7). A final section follows the stack into a rendered world and finds the sim-to-real bridge decomposes cleanly: the lookout needs a mesh-level human, the captain’s perception needs nothing, and the doors need a recipe — mixed-domain training at twice the steps — which dominates the original at the onboard scale and does not transfer down or up unchanged (§9).

Every one of those numbers comes from a cell that was registered before it ran: the question, the arms, the metrics, the reading bands with their numeric boundaries, the admissibility rule, the seeds, and, from the fourth rung on, explicit predictions with numbers. The predictions are scored afterwards, the wrong ones included, and each result names its obvious follow-ups without registering them. The protocol is the second contribution, and it is what makes the first one falsifiable: “the bus carries the mode” is a sentence with a band, not a caption on a table. §2 sets out both the architecture and the protocol; §3–§7 present the five claims in the order the evidence was built; §8 places the work; §9 states the boundaries and the transfer result.

2 Method

The bus and the door. The conditioning bus is the program’s standing instrument [1, 2]: a small hypernetwork (the *door*) maps a feature vector to a packet — a 256-dimensional latent over 16 positions — and a set of receivers add a low-rank projection of that packet into the residual

stream of a frozen language model at fixed depths, scaled by a dose. The frozen weights never change; a door with dose zero is bit-identical to the base model with every hook armed, and that identity is a gate under every cell in this paper. On a vision-language host the injection goes into the language tower after the vision tower has done its work: image tokens are already in the context when the packet lands, and the packet is injected from the last prompt position. Depths follow the registered depth-fraction prior from the 27B text-side work: the source tap at $\text{round}(0.79 \times \text{depth})$ and receivers at $\text{round}(\{0.29, 0.46, 0.63\} \times \text{depth})$ of the language tower, clamped and deduplicated, with no layer scan (the scan on the 27B could not discriminate; the prior is the stated caveat).

Mode doors. A mode is delivered as a packet computed from *mode text alone*. The feature is the host’s own tap: the last-position hidden state, at the source layer, of a two-turn text-only exchange — a mode-assignment phrasing from the user (“Be sociable: greet any person you see.”), “Understood.” from the assistant. Ten phrasings per mode, eight for training and two held out; every packet read at evaluation comes from a held-out phrasing. Two doors are trained per seed. The MODE door sees a training phrasing’s tap of a uniformly drawn mode and is teacher-forced, through the full VL forward with a uniformly drawn training frame’s image tokens in place, toward the oracle action for that mode on that frame, with a three-way auxiliary head on the packet. The CONSTANT door sees the mean training tap and is trained toward the union of the three oracles over the training frames — the habit, what the host does with no mode. Both use the same recipe throughout: 192 steps at batch 8, AdamW at 3×10^{-3} , gradient clip 1.0, dose 0.15; the “ $2\times$ ” doors of §5 and §9 change only the step count, and the “mixed” doors of §9 only the training frames. The door never sees an evaluation frame, and the packet is frame-blind by construction: it is a function of the mode phrasing and nothing else, which is what lets rung 3 ask whether a door trained on photographs transfers to a rendered world without touching the door. Gates per door: held-out-phrasing token accuracy on the mode door (≥ 0.90 registered; a weak gate by shape, since most of those tokens are JSON scaffolding), zero-dose bit-exactness with every hook armed, and byte-identical parameter samples of the frozen weights at every boundary. Coherence is read from the parse-fallback share of each cell, never halved away. Three door seeds, 42 / 1042 / 2042; the frames do not change with the seed.

Hosts. LFM2.5-VL-450M and LFM2.5-VL-3B (the creature and onboard scales; 16-layer hybrid language towers, in-harness **transformers**) and Qwen3.8-27B with its vision tower — the same 27B the text-side goal-pursuit work conditioned, cross-referenced rather than retrained. All decoding is greedy with a 24-token budget; the answer is parsed as the first JSON object carrying an action in the vocabulary, else the first vocabulary word in the decode, else a recorded fallback. Everything runs on one RTX PRO 6000; the 27B is streamed to CUDA in BF16 (51 GiB resident, 70 GiB peak through door training) under a service block that stops the production models it displaces and restores them from an exit trap.

The lookout and the switch. The small model is a person classifier: a MobileNetV3-small head in rung 2 and COCO-pretrained YOLO26 detectors of three sizes from rung 2y on, the detector reading the same PRESENT rule (a person box $\geq 5\%$ of the frame) off its own boxes. The switch is the whole coupling: the verdict selects which packet is on the bus — the person-mode’s on “present”, explore’s on “absent” — and nothing else passes between the two models. Rung 2b adds a learned projection of the detector’s boxes onto the bus as the one exception, and §6 reports what it did. The latch and gates of §7 sit in front of the switch: they decide whether the large model runs at all on a given frame.

The registration protocol. Every cell in this paper was registered in the track plan before it ran: the question, the world, the arms, the metrics, the reading bands — (i) through (v), each a named outcome with a numeric boundary — the admissibility rule that disqualifies a row from being read, the seeds, the venue, and, from rung 2y on, explicit predictions with numbers. Readings are taken by the majority of the three seeds, with the pooled number recorded beside the majority; where the rule’s verdict and the honest reading disagree (rung 1b’s asymmetry, where one mode drops out of the majority for having no band), both are printed. Predictions are scored after the fact including the wrong ones — the detector-size prediction was pessimistic by three frames, the 27B repair prediction was moot, the 27B rung 2y false-alarm band was missed by 0.02 on one door — and every registration names, without registering, the follow-ups it makes obvious, so the next cell is a matter of record before its result can tempt it. The protocol is the methods contribution alongside the architecture: the bands are what make “the bus carries the mode” a falsifiable sentence rather than a description of a table.

3 The bus carries modes

Rung 1 asks the simplest question the architecture has: does a vision-language host take a mode from the bus *while it is looking at a frame*, and does it keep looking while it does? Every bus result before this one conditioned a text-only exchange. Image tokens are an input channel the packet had never had to coexist with, and the registered failure reading was exactly that coexistence failing: the prompt door works, the bus does not, once the frame is in the context.

The world is one camera frame and one fixed question — *You are a mobile robot. This is your camera frame. What do you do next?* — with a three-word answer vocabulary: **greet**, **stop**, **proceed**. Three modes map one-to-one onto those actions: **social** greets a person and proceeds past an empty room, **caution** stops for a person and proceeds otherwise, **explore** proceeds always. Frames are COCO val2017; PRESENT means at least one non-crowd person box covering $\geq 5\%$ of the image, ABSENT means no person annotation at all; 30+30 evaluation frames, disjoint from the 40+40 the doors were trained against, the same frames under every door seed and every host. Each mode has ten assignment phrasings, eight for training and two held out; every packet the evaluation reads was computed from a held-out phrasing. The doors themselves are the recipe of §2: a mode door and a constant (habit) door per seed, 192 steps at batch 8, dose 0.15, the packet computed from the mode text alone and injected at three depth-fraction layers of the language tower while the frame’s image tokens sit in the context.

Four arms per mode on the person frames. The *packet* arm reads the mode from the bus. The *shuffled* arm injects the *next* mode’s packet in a fixed cycle — the swap control, which separates “a packet moved the answer” from “*this* packet moved it toward *its* action”. The *habit* arm injects only the constant door: what the host does with no mode at all. The *text* arm prepends the same held-out phrasing to the user turn with the constant door armed — the mode delivered through the prompt, the channel the bus is competing with. Two preregistered numbers carry the reading: LEAN = packet minus habit share of the mode’s action, and TEXT LEAN, the same for the prompt. On the empty frames, one more: GROUNDING, the share of the mode’s action when it is the wrong action — greeting nobody, stopping for nobody. Reading (i), LEAN, required packet lean ≥ 0.25 at two of three modes, packet lean $\geq 0.7 \times$ text lean, and grounding ≤ 0.20 ; reading (ii), POSSESSION, was the creature-scale signature the goal-pursuit line had seen at 230M — a mode that owns the answer regardless of the frame; reading (iii), NULL, the coexistence failure. Each host is read by the majority of three door seeds.

Table 1 is the result at three hosts. At the 27B every mode is alive and the packet beats the

prompt sentence in all three: leans 0.64/0.87/0.47 against text leans 0.53/0.48/0.13, the mode’s action reaching 0.98/1.00/1.00 of person frames under the packet where the same words in the prompt reach 0.87/0.61/0.67. Reading (i) at all three seeds. The swap control is clean — the next mode’s packet gives the action 0.00–0.01 — so the packets are mode-specific, not a generic nudge. At the 3B two modes carry at prompt strength (caution lean 0.67 vs text 0.82; explore 0.56 vs 0.52, the packet ahead) and one is dead: the **social** packet leans 0.02 against a text lean of 0.46. It does not carry “greet”; it moves the habit’s proceed into stop, a weak caution packet wearing the social phrasing. Reading (i) by the registered letter, two of three seeds, with the failing mode named. The 450M reads MIXED: caution at or above the prompt at every seed (0.78 vs 0.70), social at half the prompt’s strength (0.33 vs 0.67), and explore with no lever on either channel (0.04 vs 0.06) because the habit already proceeds half the time and the instruction cannot raise it. Direction is carried even where the lean is short: the swap number is +0.49 to +0.80 at the 3B and +0.52 to +0.68 at the 450M — the caution packet never greets, the social packet never stops more than it greets.

The 27B settles what the 3B’s dead channel was. The same door recipe, the same three phrasings, the same frames, and **social** leans 0.43/0.53/0.97 by seed at the large host. The greet direction was the small host’s weak channel, not the door’s, and paper 17’s ladder shape — the lean appears with scale — holds across three hosts of one architecture: one mode at parity at 450M, two at 3B, three above the prompt at 27B.

The number the architecture was built for is the last column. On the empty frames the packet’s wrong action is ≤ 0.04 pooled at every host (3B 0.04/0.02, 450M 0.01/0.03, 27B 0.00/0.04; one 27B door, s2042, reaches 0.13 for caution), and **proceed** is 1.00 under every packet. There is no possession at any scale. The prompt is the channel that loses the frame: the same phrasing in the user turn makes the 3B greet an empty room 0.48 of the time and stop for nobody 0.58; the 450M 0.21 and 0.38. The instruction overrides what the model sees; the bus mode leans on what it sees. Both channels reach the model, but only one of them arrives *as a way of looking* rather than as a fact to be acted on. In plain terms, honestly flagged as an analogy: written orders make the guard say hello to an empty doorway half the time; the mood does not.

Two gates ran under every cell and are reported for completeness: with every hook armed and dose zero, the decode is bit-identical to the base model, and parameter samples of the frozen weights are byte-identical at every boundary. Held-out-phrasing token accuracy on the mode door read 0.97–0.99 at the LFM hosts and 1.00 at the 27B — a weak gate by shape, since most of those tokens are JSON scaffolding, recorded as such. Zero parse fallbacks in 2,520 LFM decodes and in every 27B cell.

host	mode/action	packet	shuffled	habit	text	empty wrong	packet seeds
450M	social (greet)	0.63	0.00	0.30	0.97	0.01	0.53/0.87/0.50
450M	caution (stop)	1.00	0.32	0.22	0.92	0.03	1.00/1.00/1.00
450M	explore (proceed)	0.52	0.00	0.48	0.53	–	0.53/0.53/0.50
3B	social (greet)	0.49	0.00	0.47	0.92	0.04	0.50/0.50/0.47
3B	caution (stop)	0.76	0.00	0.09	0.91	0.02	0.27/1.00/1.00
3B	explore (proceed)	1.00	0.20	0.44	0.97	–	1.00/1.00/1.00
27B	social (greet)	0.98	0.00	0.33	0.87	0.00	1.00/0.97/0.97
27B	caution (stop)	1.00	0.00	0.13	0.61	0.04	1.00/1.00/1.00
27B	explore (proceed)	1.00	0.01	0.53	0.67	–	1.00/1.00/1.00

Table 1: Rung 1 on person frames. Shares are pooled over seeds; the final column is seed 42/1042/2042. Empty wrong is the packet action on absent frames and is undefined for explore.

4 The switch is oracle-equivalent

Rung 1 chose the packet from the dataset label. The architecture has a *small* model look at the frame fast and emit the mode; the large model looks for itself. Two questions follow. When the small model is right, does the system behave as it did under the oracle? When it is wrong, which model owns the behaviour — the one that emitted the mode, or the one that can see?

The construction is a switch, deliberately: the lookout’s verdict (person / no person) selects rung 1’s packet, and nothing about the doors changes. An operator picks a PERSON-MODE $M \in \{\text{social}, \text{caution}\}$; verdict *present* sends M ’s packet, verdict *absent* sends **explore**’s (“nobody here, keep going”). Frames are 100 present and 100 absent, drawn by the fixed generator after rung 1’s draw and disjoint from it; the doors never saw any of them. Four arms per mode, host, and seed: ORACLE, the label drives the policy; CLASSIFIER, the lookout’s verdict drives it; FLIPPED, the verdict inverted on every frame — the wrong-lookout cell at full size, independent of the real error rate; and HABIT. Three preregistered numbers: FIDELITY = classifier minus oracle share of M ’s action on person frames; the FALSE-ALARM cell, M ’s action share on *empty* frames under FLIPPED (the lookout cries “person”, the room is empty); and the MISS cost, oracle minus FLIPPED on person frames (the lookout says “nobody”, a person is there). Reading (i), ROBUST, required $|\text{fidelity}| \leq 0.10$ and $\text{false-alarm} \leq 0.20$ for both modes; reading (ii), CAPTURED, was $\text{false-alarm} > 0.50$ — a false alarm from the small model owning the big model’s answer.

The lookout was changed three times without retraining a door, and Table 2 is the ladder. Rung 2 used a MobileNetV3-small head (2.5M parameters, about 1 ms a frame), a linear layer on frozen ImageNet features trained for three epochs on COCO frames outside every evaluation set: 0.91 on the 200 (6 false alarms, 12 misses). ROBUST at both hosts (450M three of three, 3B two of three), fidelity $-0.05/ -0.08$ at the 450M and $-0.04/ -0.07$ at the 3B, false-alarm cells 0.01/0.01 and 0.02/0.04. Rung 2y put a COCO-pretrained person detector in the same seat with zero parameters trained, reading rung 1’s PRESENT rule off its own boxes: yolo26n scores 0.965 on the 200 (2 false alarms, 5 misses) and 0.986 on 3,472 calibration frames. ROBUST three of three at both hosts; fidelity tightens to $-0.03/ -0.04$ and $-0.01/ -0.03$; the 3B’s false-alarm cell reproduces rung 2’s *to the decimal*, 0.02/0.04. Then the size axis: yolo26s and yolo26m (9.5M and 20.4M parameters, 3.6 and 4.4 ms) find every one of the nano’s five misses as a confident, large box (0.34–0.90 confidence, 11–97% of the frame; Table 3) — the nano simply did not see them, a recall floor rather than a threshold — and with zero misses the classifier arm *is* the oracle arm on person frames: fidelity 0.00 at every seed of both hosts, the 3B’s false-alarm cell 0.02/0.04 for the third time under a third small model. The registered rule (fewest calibration misses, ties to the smaller) picked the medium; the operating detector for later rungs is the small, which has no misses and one fewer false alarm on the 200 and yields pooled cells identical to the medium’s. At the 27B, with the nano, ROBUST three of three: oracle / classifier 0.94/0.90 and 0.97/0.92, fidelity $-0.03/ -0.05$, false-alarm 0.00/0.07 (one caution door, s2042, at 0.14 — the door that grounds 0.13 in rung 1).

Every fidelity gap in that ladder is the same product. A miss silences the mode by construction (**explore** means proceed), so the classifier arm can only fall short of the oracle by the miss rate times what a missed person costs, and it does: rung 2’s worst row, the 3B’s s2042 caution at -0.12 under a 12-miss head, is -0.05 under the 5-miss nano — $5\% \times$ that seed’s 0.94 miss cost — and 0.00 under the miss-free small and medium. At the 27B the miss cost is 0.90/0.92 because the door delivers the whole mode, and $-0.03/ -0.05$ is 5% of that. The bound held across three lookouts and three hosts with the same doors and the same frames; it is the switch’s arithmetic, not any classifier’s. The rule for choosing a lookout follows from it: recall is the number to buy, and a detector that stops missing buys the oracle’s behaviour outright.

The false-alarm side is where the authority lives, and it is best read frame by frame. The nano’s

two “false alarms” on the 200 are not detector noise. Image 261706 is a cat sprawled on a couch holding a television remote (person 0.41, 34% of the frame — a fair mistake). Image 509699 is a woman in an armchair whom COCO never annotated; the detector is right and the label is wrong, at every size. On those two frames the 27B, at every seed and under both modes, says **proceed** to the cat — overruling the lookout — and **greet** or **stop** to the woman, agreeing with the lookout against the dataset. The LFM hosts split the same way in aggregate (the mode fires 0.50/0.33 at the 450M and 0.00/0.33 at the 3B across the six rows), and the small detector’s one disagreement with the medium is exactly the cat, which is why their pooled cells are identical: the host ignores the packet on that frame under either verdict. On the misses the picture is the mirror image: the mode fires 0.00 on all fifteen rows at the 3B and the 27B — the host obeys the lookout’s silence completely, because a mode that was never sent cannot be overruled. The big model is the authority on the frame in both directions: it corrects the small model’s one real mistake and backs its one correct call that the dataset got wrong.

That is the division of labour the architecture wanted, and it is not a property of any lookout. Swap the small model — a trained head, a pretrained detector, three detector sizes — and the large model’s behaviour on the frames the lookout gets wrong is reproduced to two decimals while the switch’s fidelity tracks the miss rate exactly. In plain terms: the lookout on the mast changed three times and the captain behaved the same each time, following the lookout when the lookout was right, trusting their own eyes when the lookout was wrong; hire a lookout who never misses and the captain behaves as if handed the crew manifest — and still ignores the lookout about the cat.

host	lookout	mode	fidelity seeds	false alarm	miss	miss cost
450M	head	social	-0.07/-0.03/-0.06	0.01	0.19	0.42
450M	head	caution	-0.08/-0.09/-0.06	0.01	0.34	0.63
3B	head	social	-0.04/-0.03/-0.04	0.02	0.04	0.42
3B	head	caution	-0.01/-0.09/-0.12	0.04	0.07	0.66
450M	yolo26n	social	-0.03/-0.02/-0.03	0.02	0.17	0.45
450M	yolo26n	caution	-0.04/-0.03/-0.04	0.01	0.30	0.67
3B	yolo26n	social	-0.02/0.00/-0.02	0.02	0.01	0.44
3B	yolo26n	caution	0.00/-0.05/-0.05	0.04	0.03	0.70
450M	yolo26s	social	0.00/0.00/0.00	0.02	0.14	0.48
450M	yolo26s	caution	0.00/0.00/0.00	0.01	0.26	0.71
3B	yolo26s	social	0.00/0.00/0.00	0.02	0.00	0.46
3B	yolo26s	caution	0.00/0.00/0.00	0.04	0.00	0.74
450M	yolo26m	social	0.00/0.00/0.00	0.02	0.14	0.48
450M	yolo26m	caution	0.00/0.00/0.00	0.01	0.26	0.71
3B	yolo26m	social	0.00/0.00/0.00	0.02	0.00	0.46
3B	yolo26m	caution	0.00/0.00/0.00	0.04	0.00	0.74
27B	yolo26n	social	-0.05/-0.03/-0.02	0.00	0.03	0.90
27B	yolo26n	caution	-0.05/-0.04/-0.05	0.07	0.05	0.92

Table 2: Fidelity ladder. Fidelity is classifier minus oracle action share per seed; the remaining cells are pooled.

detector	parameters	accuracy	misses	false alarms	ms/frame
yolo26n	2.4M	0.965	5	2	5.6
yolo26s	9.5M	0.995	0	1	3.9
yolo26m	20.4M	0.990	0	2	4.8

Table 3: Off-the-shelf detector size sweep on the 200-frame evaluation set.

5 Robustness and its price

A mode on the bus is only worth having if it survives the prompt. Rung 1b stages the contest directly: the client’s text argues against the operator’s mode while the model looks at a person. The hostile phrasing for a mode is its *opponent’s* held-out phrasing — **caution** is attacked by “proceed no matter what”, **explore** by the caution text, **social** by explore’s — and the congruent phrasing is the mode’s own. Six arms per mode, the goal-pursuit line’s: packet alone; hostile text alone (constant door plus the opponent’s sentence, the text arm of §3 turned against the mode); packet plus hostile; packet plus congruent; habit; and shuffled plus hostile, the opponent’s packet under the opponent’s sentence. The preregistered number is OWNERSHIP,

$$\text{ownership} = \frac{\text{share}(\text{packet} + \text{hostile}) - \text{share}(\text{hostile})}{\text{share}(\text{packet}) - \text{share}(\text{habit})},$$

the fraction of the mode’s own lean that survives a direct contrary order, pooled over seeds, with a 0.05 denominator guard and read only on the modes that have a lean at that host. A (host, dose) row is ADMISSIBLE only if parse fallbacks are ≤ 0.05 and the packet’s wrong action on empty frames is ≤ 0.20 — the row must keep the frame to count. Reading (i), PACKET HOLDS, was ownership ≥ 0.7 at two of three seeds; (ii) SPLIT, 0.3–0.7; (iii) TEXT WINS, below 0.3; (iv) ASYMMETRY, readings that differ by mode, recorded per mode. Two dose rows, each its own artifact: $\times 1$, the trained dose of 0.15, and $\times 1.5$, the operating number the text-side dose scan had found on the same 27B.

Table 4 is the contest at three hosts. At the 3B and the trained dose the honest reading is the asymmetry. **Explore** against the caution text holds outright: the text alone drives proceed from the habit’s 0.44 down to 0.06 — a hostile with teeth — and packet plus hostile keeps it at 0.92; ownership 1.56 pooled, three of three (1.04/2.30/1.93). **Caution** against the explore text splits: the text alone drives stop to 0.00, the packet alone gives 0.76, packet plus hostile 0.41; ownership 0.62 pooled, and by seed 0.00/0.48/0.83 — ordered exactly by rung 1’s caution lean at those seeds, 0.13/0.90/0.97. The stronger the door, the more of it survives. That ordering is the hinge of this section and is returned to below. The 450M holds (0.80/1.04) with an asterisk recorded at the time: its opponent text is explore’s phrasing, which has no lever at that host (rung 1 text lean 0.06), so the packet is undisturbed by a prompt that could not have moved the model anyway — a weak adversary. At the 27B, where the door is the stronger channel to begin with, PACKET HOLDS three of three: ownership 0.84/0.90/1.48 for social / caution / explore, the caution text alone pulling proceed down to 0.29 (teeth) and the packet shrugging it off. By seed 1.46/1.17/1.08 and 1.25/0.92/0.83, with s2042 the odd door again — social 0.34, caution 0.67, the same door whose grounding pulses 0.13 in rung 1. Its strength leaks as suggestibility; door temperament, not just door strength, varies by seed.

The packet does more than survive at the 3B: it repairs the prompt. On empty frames the caution text alone makes the 3B refuse to proceed 0.67 of the time — it stops or greets nobody, the ungroundedness of §3 under adversarial conditions — and with the explore packet under that same text the model proceeds on 0.99. The operator’s mode does not merely outlast the client’s contrary order; it restores the model’s grip on what it sees. The 27B needs no such repair: the hostile text alone does not move it on empty frames (wrong action 0.00, proceed 1.00 under the caution text), so the prediction that the packet would repair the prompt’s hallucination was moot in the best way. The biggest host grounds its own prompt.

Then the two levers, and their prices. The first is dose. At the 3B, $\times 1.5$ buys caution ownership 0.84 (0.38/0.93/1.08) and loses the frame: the caution packet stops for nobody on 0.43 of empty frames at s2042, the social packet greets the empty room 0.27 at s42, and the constant door becomes

a greet-compulsion (habit greet 0.99). Admissible pooled by the letter, inadmissible at two of three seeds; the 450M is inadmissible outright at $\times 1.5$ (fallbacks 0.17 at s2042). At the 27B the same dose is free: ownership 0.91/0.93/0.89, grounding 0.00/0.03, zero fallbacks, and even s2042 lifts (social 0.34 $\rightarrow$ 0.70, caution 0.67 $\rightarrow$ 0.73) at no grounding cost. The dose window splits by *host*, not by modality — the text-side scan on the same 27B had found $\times 1.5$ free and $\times 2$ fatal, and the VL side reproduces the first half of that — and it splits one notch earlier at 3B than at 27B. Scale buys the safety margin that lets the dial move.

The second lever is door strength, and the seed ordering said it costs nothing. The stronger-door cell tested it where it was weakest: the 3B s42 door pair retrained with the same recipe, generator, and frames at $2\times$ the steps (192 $\rightarrow$ 384), saved beside the registered doors, never overwriting them, and read against the registered s42 row (caution ownership 0.00, packet share 0.27). Reading (i) required ownership ≥ 0.48 — overtaking the registered middle seed — *and* admissibility *and* grounding within 0.05 of the registered row. The pair trained in 184 s. Caution packet share 0.27 $\rightarrow$ 1.00, packet plus hostile 0.00 $\rightarrow$ 0.73, ownership 0.00 $\rightarrow$ 1.47 — past every registered seed — with zero fallbacks and grounding 0.00 (Table 5). Explore holds (1.11). Social did not lift (0.50 at both strengths): the greet channel’s ceiling at this host is not steps either. Where $\times 1.5$ dose bought 0.84 ownership at the price of the frame, $2\times$ steps bought 1.47 for nothing. The asymmetry between the two levers is the section’s result: dose and strength both raise ownership, but only dose spends grounding, because dose scales what the packet writes into the residual stream on every frame while steps sharpen what the packet *is*. Rung 3 (§9) will find the same lever paying for a second thing, domain transfer, at the same zero price.

The emitter was not the variable. The 1-ms classifier head and the person detector, put in the oracle’s seat in the contested cells, reproduce every oracle row within their misses: packet shares -0.03 to -0.06 , ownership unchanged to two decimals (450M $\times 1$ 0.81/1.03 against 0.80/1.04; 3B $\times 1$ caution 0.64 against 0.62, explore 1.56 exactly; the detector, one error on 60 frames). And because a correct “absent” verdict keeps empty frames on explore’s packet, the lookout guards grounding as a side effect.

In plain terms, flagged as an analogy: tell the 3B in writing “stop for people” while its mode says “keep going”, and it keeps going 92% of the time; tell it “keep going” while its mode says “stop for people”, and it stops 41% of the time, more where the mode was trained harder, and 100% once that seed’s door was trained twice as long — still without ever stopping for a ghost. Turn the mode’s volume up by half instead and it also stops 85% of the time, but now it starts stopping for nobody. At the 27B the mode wins about nine times in ten against a direct contrary order at either volume, and never stops for a ghost.

host	dose	mode	ownership seeds	pooled	grounding
450M	x1	caution	0.97/1.33/1.00	1.04	0.03
450M	x1	explore	-/0.90/-2.80	-	-
450M	x1.5	caution	1.00/3.00/1.00	1.06	0.08
450M	x1.5	explore	0.54/1.25/0.08	-0.11	-
3B	x1	caution	0.00/0.48/0.83	0.62	0.02
3B	x1	explore	1.93/1.04/2.30	1.56	-
3B	x1.5	caution	0.38/0.93/1.07	0.84	0.18
3B	x1.5	explore	1.00/1.00/0.97	0.99	-
27B	x1	caution	1.17/0.92/0.67	0.90	0.04
27B	x1	explore	1.08/0.83/-	1.48	-
27B	x1.5	caution	1.08/1.00/0.73	0.93	0.03
27B	x1.5	explore	0.93/0.33/-	0.89	-

Table 4: Oracle-emitter ownership against hostile prompt text at the registered and 1.5x doses. Seed order is 42/1042/2042.

host	emitter	ownership	packet+hostile	grounding
450M	oracle	1.04	0.99	0.03
450M	classifier	1.03	0.92	0.00
450M	detector	1.04	0.99	0.00
3B	oracle	0.62	0.41	0.02
3B	classifier	0.64	0.40	0.00
3B	detector	0.62	0.41	0.00
3B	oracle, 2x-step door (s42)	1.47	0.73	0.00

Table 5: Emitter reproduction at dose 1x, plus the command-strength 3B s42 door trained for twice the steps.

6 Where the safety lives

Every grounding result so far — false alarms nearly free, the prompt’s ghosts repaired, the lookout overruled about the cat — was measured on a host that could see the frame. Two cells locate the property. The first widens the bus and asks whether more than one bit should cross it; the second blinds the captain and asks what the bus does when it is the only eyes.

Boxes on the bus. Rungs 2 and 2y are a switch: the lookout’s one bit picks one of the operator’s pre-trained packets, and the policy is written by hand. Rung 2b puts the lookout’s *output* on the bus instead. Per frame the detector yields an eight-number box vector — presence by the registered rule, box count, the largest box’s area fraction and confidence, its centre and size — all zeros when no person box exists. A learned projection of that vector is added to rung 1’s mode tap, in the door’s normalized source space, and the sum passes through rung 1’s mode door, loaded and frozen, to the three inject layers. Only the projection trains ($8 \times$ hidden parameters, zero-initialized so that step 0 *is* rung 1’s door), toward the oracle action of (mode, label) on rung 1’s training frames — “social plus no boxes, proceed” is learned, not written. Four arms per person-mode on rung 2’s 200 frames: BOXES, the bridge on the frame’s real boxes; SWITCH, rung 2y’s classifier arm decoded on the same path; LYING BOXES, where a present frame’s box vector is pasted onto each empty frame and zeros onto each person frame — the lookout lying in both directions through the channel; and HABIT. GAIN is boxes minus switch on person frames; the LIE cell is the mode’s action on empty frames under lying boxes; the SILENCE cost is what zeros on a person frame take away. Reading (i), BOXES HELP, needed gain ≥ 0.10 for both modes with a lie cell ≤ 0.20 ; (ii), CHANNEL OWNS, a lie cell above 0.50 — the small model’s lie capturing the host, the failure the switch had been shown not to have; (iii), ONE BIT, $|\text{gain}| \leq 0.10$ with the lie cell clean.

An amendment is on the record. The first attempt added the projection to the raw tap at rung 1’s learning rate; the door’s fitted source scale is 0.06 per dimension, the projection drifted to norm 4.9 — a shift two and a half times the whole tap spread — and the training loss *rose* from its floor, because the objective was already minimized before the boxes arrived. That is a harness artefact and was recorded as one; the rerun works in the normalized space at a tenth of the learning rate, and the projection stays small (norm 0.27–1.5). The door-alone loss at step 1 already sits at 10^{-4} to 0.08, and the final loss matches it: the objective is at its floor before the boxes arrive, which is the result in miniature.

The result is Table 6. The 450M reads ONE BIT pooled: social 0.53 under boxes against 0.59 under the switch, caution 0.92 against 0.93, gains -0.06 and -0.01 , and on two seeds the boxes *cost* the weak social door up to 0.29. The 3B reads ONE BIT at two seeds and MIXED at the third, and the mix is the finding: caution under boxes 0.93 against the switch’s 0.70, gain $+0.22$ pooled — all of it from s42, the weakest caution door in the study (rung 1 lean 0.13), where the switch delivers 0.26 and the boxes 0.83. At the strong doors the gain is $+0.05$. The channel never owns the host: the lie cell is ≤ 0.06 at every seed of both hosts. A person’s boxes pasted onto an empty frame move the big model at most six percent of the time, and zeros on a person frame merely silence the mode, at a cost within a tenth of the switch’s miss cost, as predicted: a lying “nobody” and a missed person are the same silence. The honest surprise is that a learned projection from detector boxes can compensate for a weak door ($+0.57$ stop at the 3B’s s42). The box channel is a door-strength top-up, not an information channel; where the door already delivers the mode it has nothing to add, and where the boxes lie the host looks at the frame and ignores them. The switch remains the architecture: the same behaviour, no training, and no new surface for the small model to capture the large one through.

The blind captain. Rung 2c keeps rung 2b’s harness and replaces the host’s image with a registered blank — a uniform black 640×480 frame, sha-recorded, used for every training and evaluation row — while the detector sees the real frames and supplies verdicts and boxes. The boxes are now the only eyes. One new number: DELIVERY, the person-mode’s action share on present frames minus on absent frames — the bus’s discrimination with nothing to check it against. Reading (i), BUS CARRIES THE FACT, needed switch delivery ≥ 0.50 for both modes; (ii), BUS TOO WEAK, below 0.25 for either. CAPTURE is recorded beside the reading rather than read: a lying absent-share ≥ 0.50 means the blind host obeys the lie, expected, and the confirmation that sight was the safety. The registered prediction was (i) at the 3B — the packet had moved the sighted 3B at 0.7–1.0 and, blind, nothing would oppose it.

The prediction was wrong in the interesting direction (Table 6). Blind the host and the switch delivers nothing at the trained dose: under every rung 1 packet both hosts answer **proceed** on the blank frame — the 450M proceeds 1.00 under the very caution packet that moves it to 0.97 stop when it can see. Reading (ii) three of three at both hosts. The doors are *leans* — “if a person, lean stop” — and with no frame to support the antecedent the blank-frame prior wins. The trained blind projection fits under teacher forcing (loss 0.01–0.02 at the strong-door seeds) and still cannot flip the generated action; the one partial exception at the 450M (boxes delivery 0.26/0.48) has the lie mirroring it at 0.22/0.49, because a fact-injector is direction-blind by construction. Then the exception that proves both rules. The 3B’s s2042 caution door — the strongest in the study, rung 1 lean 0.97, the only door with a grounding pulse even when sighted — does cross: switch delivery 0.93 on the blind host, and on the same mode the lying boxes capture 1.00. What crosses the bus without eyes is obeyed without question.

That locates the property. A mode packet cannot make a host hallucinate, sighted or blind, because it carries a conditional lean rather than a fact; but a door strong enough to act as a command channel turns a blind host into the lookout’s puppet, and the same strength that bought robustness in §5 is what makes the capture total. Sight, not the bus, was every grounding result in §3–§4. The deployment rule follows directly: doors below command strength for any host that can lose its eyes, and keep the captain sighted, because grounding lives in its own perception and nowhere else. In plain terms: a lookout can set the captain’s mood but cannot tell the captain what is on the water — unless the captain has been blindfolded and the lookout shouts loudly enough, in which case the captain will steer wherever the lookout says, including onto the rocks.

host	captain	mode	lie/crossing seeds	top-up/capture seeds	pooled ground/lie
450M	sighted	social	0.12/0.01/0.01	0.09/-0.29/0.03	0.02
450M	blind	social	0.00/0.00/0.00	0.22/0.00/0.00	0.07
450M	sighted	caution	0.07/0.00/0.01	-0.11/0.05/0.04	0.02
450M	blind	caution	0.00/0.00/0.00	0.49/0.00/0.00	0.16
3B	sighted	social	0.01/0.02/0.02	0.02/0.00/0.01	0.02
3B	blind	social	0.00/0.00/0.00	0.00/0.00/0.00	0.00
3B	sighted	caution	0.03/0.04/0.06	0.57/0.05/0.05	0.05
3B	blind	caution	0.00/0.00/0.93	0.00/0.00/1.00	0.33

Table 6: Rung 2b sighted box-channel cells and rung 2c blind-captain cells, seed order 42/1042/2042.

7 The drowsy captain

The big model has decoded every frame so far. The lookout costs a few milliseconds; the captain costs tens of milliseconds at the toy scale and about a second at 27B. Rung 2t asks the compute question: if the big model re-evaluates only when the lookout says the world changed, and its last decision stands otherwise, how much is saved and what does staleness cost? The limit is known in advance. A gate can preserve decisions only as far as the world, *seen through the gate*, determines them; whatever the big model decides from frame content the gate cannot see is lost while the gate sleeps.

The world becomes a stream: 60 episodes of 16 steps drawn once from rung 2’s 200 frames (seed 2026, identical for every host and door seed). Step 0 is a cut; each later step cuts to a uniformly drawn frame with probability 0.2 or stays. Every step, cuts included, is re-rendered with camera jitter — a random 94–100% crop resized back and a brightness factor in $[0.96, 1.04]$ — so naive pixel equality is dead and a stay-step is never the same file. Uniform cuts mean about half of them keep the presence label; those same-label cuts are the staleness probe, the world changing while the verdict does not. The system under test is rung 2y’s switch with a latch: at a wake the operating detector’s verdict picks the packet, the big model decodes the current jittered frame with it, and the action is latched until the next wake. Between wakes only the lookout runs. Six gates per mode, host, and seed: ALWAYS, wake every step, the baseline; NEVER, wake at step 0 only, the floor; VERDICT, wake when the detector’s verdict differs from the verdict at the last wake; BOX, wake additionally on a change in person count, a shift of more than 0.10 in the largest box’s area or 0.15 in its centre; PIXEL, the understanding-free control, wake when the mean absolute pixel difference to the previous step exceeds 0.04; and CUT-ORACLE, wake exactly on true cuts, the ceiling — its residual disagreement with ALWAYS is the big model’s own jitter-instability, and no gate can beat it. Four numbers per gate: AGREEMENT with ALWAYS’s action at each step, WAKE FRACTION (compute saved is one minus it), LAG from a true cut to the next wake, and FALSE WAKES. Reading (i), FREE LUNCH, on the VERDICT gate required agreement $\geq \min(0.95, \text{cut-oracle} - 0.02)$ and wake fraction ≤ 0.40 for both modes; (ii), STALE, agreement below 0.85; (iii), JITTERY, wake fraction above 0.70.

The ladder of gates behaved exactly as registered (Table 7, Figure 1). The VERDICT gate wakes the big model on 0.165 of steps — 83.5% of decodes saved — at agreement 0.97/0.99 (450M, social / caution) and 0.96/0.99 (3B) with the every-frame baseline; oracle correctness, the absolute quality against the frame labels, is the baseline’s within a point (450M caution 0.970 against ALWAYS’s 0.978; 3B 0.869 against 0.874). Mean lag from a true cut to the wake is 0.98 steps; false wakes are 30 of 5,760 gate-steps, the predicted flicker of borderline boxes under jitter. FREE LUNCH at both hosts, two of three seeds each (s1042 dips to 0.947 on social against the 0.95 band). NEVER, the floor, sits at 0.65–0.74: a latched first impression is wrong a third of the time by an episode’s end. PIXEL wakes on 0.947 of steps: the registered jitter’s median frame-to-frame difference is 0.107, nearly three times the threshold, so the understanding-free gate cannot tell camera noise from change and saves nothing — JITTERY, and the prediction to the letter. CUT-ORACLE reaches 0.987–0.995: the big model flips about one percent of its answers on a jittered re-render of the same frame, so no gate can score above roughly 0.99, and VERDICT sits one to three points under that ceiling. BOX buys most of the residual (0.99–1.00) for eleven points more wakes.

One registered prediction was too pessimistic and the reason is the useful part. Social was expected to pay ten to fifteen points, because its within-verdict variance is large — half the person frames greet and half do not, and the gate sleeps through the difference. It paid one to three. The gate’s error is bounded by lag times flip rate, and the lag is one step: a stale decision lives only until the next verdict change, which is never far away in a stream with cuts. The gate preserves what

presence determines and loses only what changes inside a presence run, and that loss is proportional to how long the gate sleeps through it.

The economics follow. The lookout costs 7.5 ms a frame; a skipped decode saves 45 ms at 450M and 68–76 ms at 3B, so the gate pays for itself six to ten times per skipped call at toy scale, and the ratio scales with the host — at 27B, where a decode is about a second, it is over a hundred. The speculative-decoding analogy is exact in shape and honest about its limits: a cheap model proposes “nothing changed” and the expensive model is consulted only on disagreement; unlike token-level speculation, acceptance here is not verified, and the price of a wrong acceptance is a stale action for one step. In plain terms: the robot’s big brain naps five frames out of six, the lookout wakes it within one frame of anything actually changing, and nobody watching its behaviour over 960 frames would notice more than a one-to-four percent difference — while the compute bill drops by five sixths.

host	gate	mode	saved	agreement	lag	false wakes
450M	NEVER	social	93.8%	0.646	0.00	0
450M	NEVER	caution	93.8%	0.669	0.00	0
450M	PIXEL	social	5.3%	1.000	0.00	2046
450M	PIXEL	caution	5.3%	1.000	0.00	2046
450M	VERDICT	social	83.5%	0.972	0.98	30
450M	VERDICT	caution	83.5%	0.988	0.98	30
450M	BOX	social	72.3%	0.992	0.30	225
450M	BOX	caution	72.3%	1.000	0.30	225
450M	CUT-ORACLE	social	76.4%	0.988	0.00	0
450M	CUT-ORACLE	caution	76.4%	0.994	0.00	0
3B	NEVER	social	93.8%	0.725	0.00	0
3B	NEVER	caution	93.8%	0.744	0.00	0
3B	PIXEL	social	5.3%	1.000	0.00	2046
3B	PIXEL	caution	5.3%	0.999	0.00	2046
3B	VERDICT	social	83.5%	0.964	0.98	30
3B	VERDICT	caution	83.5%	0.986	0.98	30
3B	BOX	social	72.3%	0.999	0.30	225
3B	BOX	caution	72.3%	0.997	0.30	225
3B	CUT-ORACLE	social	76.4%	0.995	0.00	0
3B	CUT-ORACLE	caution	76.4%	0.991	0.00	0

Table 7: Gate ladder pooled over seeds. The lookout costs 7.5–7.5 ms/frame; per-seed decode means are 49.8/44.8/44.0 ms at 450M and 75.9/68.3/68.1 ms at 3B.

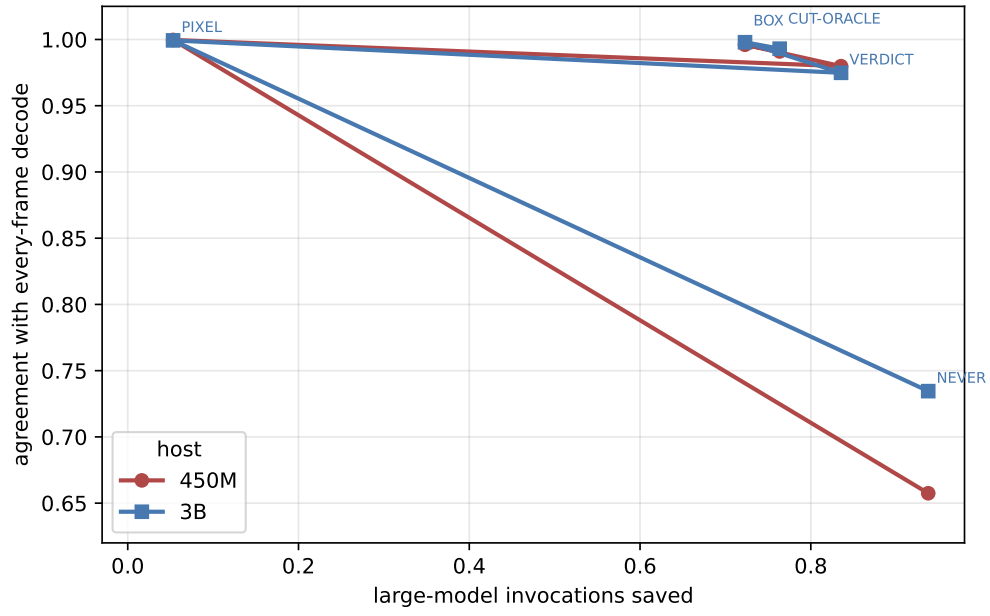


Figure 1: Saved large-model invocations against mean social/caution agreement with every-frame decoding. Labels identify the registered gates; both points and paths come from pooled rung 2t artifacts.

8 Related work

Steering a frozen model from inside. Adding a vector to a model’s residual stream to shift its behaviour is well established: activation addition [6] and representation engineering [7] derive steering directions from contrastive prompts and inject them at one or more layers, and the wider control literature has shown that the directions so found are often low-dimensional and behaviourally legible. The bus of this paper is a steering channel in that family with three differences that matter for the result. The direction is not derived from a contrast but *trained through the host* toward an action, from a feature that is itself the host’s own state on the mode text; it is generated by a hypernetwork [8] rather than stored, so the same door serves every mode phrasing including held-out ones; and it is injected on a vision-language host with image tokens in the context, which is the coexistence question §3 answers. The grounding result — a steering channel that leans on what the model sees rather than overriding it — has no direct precedent we are aware of, and the blind-captain cell shows it is a property of the door’s strength, not of steering as such.

Soft prompts, prefixes, and adapters. Prefix tuning [9] and prompt tuning [10] condition a frozen model through learned embeddings in the prompt slot, and LoRA [11] through low-rank weight deltas. The program this paper belongs to has compared those doors on the same hosts: a want enters best where language enters, through a prefix, while a manner — a way of acting — enters best through the bus [2]. The modes here are manners in that sense, which is why the bus is the channel under test rather than a prefix. The door recipe (a conditioned latent bridge with receivers at depth fractions) is the one that work established at 27B [1], unchanged on the language tower of each VL host.

Small models in front of large ones. Vision-language agents routinely place a cheap perception module ahead of a large model, and tool-use frameworks let the large model call detectors as functions; the difference here is the direction and the type of what crosses. The lookout does not return a fact into the large model’s context, where it would be one more overridable token sequence; it sets a stance through a channel the context cannot see, and the large model keeps the facts. The compute result of §7 is a behavioural analogue of speculative decoding [12, 13]: a cheap model proposes that nothing changed and the expensive model is consulted only on disagreement, with the difference that acceptance is not verified and the cost of a wrong acceptance is one stale action. The gate’s failure mode is also the classical one from change detection and event-based vision [14]: a threshold on raw pixel change cannot separate camera noise from world change, and only a gate with a semantic verdict can.

Sim-to-real for perception. The rendered-world result of §9 touches the sim-to-real literature only at its narrowest point — whether a COCO-trained detector reads a rendered person — and finds the answer is entirely about the figure’s shape and texture rather than the renderer, in line with the domain-gap work that motivates photorealistic and scanned assets. What is specific to this architecture is that the doors’ transfer, unlike the detector’s, is a training-recipe question rather than an asset question, and one with a lever that also buys robustness.

Preregistration. The registration protocol follows the practice of preregistered studies in the empirical sciences — hypotheses, analyses, and decision rules fixed before data are seen — applied cell by cell to an engineering ladder, with the additional discipline that each result names, without

registering, the follow-ups it makes obvious, so that the next cell is on record before its result can shape it.

9 Limitations and rung 3

The study’s boundaries are stated before its transfer result, because the transfer result is what tests them. Every number in §3–§7 lives on COCO photographs and, for the stream, on synthetic jitter episodes over those photographs; the action space is three teacher-forced words; the doors are one recipe on one language-tower depth prior, and the 27B is one model family. Three door seeds is enough to read a majority and to catch a headline that a replicate reverses — it has, twice in this program — and not enough to estimate a seed distribution. The registration protocol bounds what can be claimed but not what can be imagined: a band written before a run is a commitment, not a proof that the band was the right one, and the honest readings in §5 and §6 where the rule’s verdict and the numbers disagreed are the protocol working as intended rather than a flaw in it.

Rung 3 is the boundary the robot cares about: render the world instead of photographing it, and ask what transfers with nothing retrained. The architectural stake was stated in advance. A door’s packet is computed from the mode phrasing’s tap and never from the frame, so the doors cannot fail to transfer *by construction*; what is actually under test is the two perceiving parts, the lookout’s verdict and the captain’s own grounding, on rendered pixels. The scene is a floored room with two to four coloured box obstacles, a camera with position and yaw jitter, a jittered light, and a person placed within view; the present rule is rung 1’s, measured exactly instead of annotated — the person’s segmentation mask must cover at least 5% of the frame — and absent is the same drawn scene with the person removed; 100 present and 100 absent frames from one draw.

The first person was the simulator’s stock capsule humanoid, at two registered realism levels (stock appearance, then skin- and clothing-coloured geoms). Table 8 is the reading: zero person verdicts in 600 detector-level trials across three detector sizes and both levels, zero false alarms, accuracy 0.50 — a total sim gap. The mechanism is the useful part. This is not weak person-signal under the threshold; it is a confident wrong parse. On present frames the detector’s best person confidence peaks at 0.03 while the same frames yield *sports ball* at 0.81 (the head), *orange*, *carrot*, *knife*, and *baseball bat* (the capsule limbs). Capsule primitives never enter the person region of COCO feature space, whatever their colour; the registered realism ladder was aimed at the wrong axis. It is worth saying plainly for anyone evaluating COCO detectors against primitive-based simulators: zero false alarms and zero hits look like a conservative detector and are total blindness. The captain, probed on the same capsule frames under the caution text, discriminates at 0.267 — stopping for the figure 27 points more often when it is present — where every YOLO scored exactly zero: MIXED by the registered band, and directionally the captain out-sees the lookout in simulation.

One asset swap closes the gap. Rung 3b replaces the capsule with a pinned photogrammetry scan of a clothed adult (HSR0081 from HSRD-100, CC BY 4.0), same rooms, cameras, lights, and draws; the render differs from rung 3’s only in the figure. The operating detector goes from 0/100 to 99/100 present with zero false alarms, accuracy 0.995. The sim gap was the figure, entirely. Then the doors (Table 9). At the 450M the unchanged COCO doors travel at or above their photograph leans (s42 caution 1.00 against 0.78 on COCO; social 0.50 against 0.33), grounding 0.00 everywhere: for the small host the clean render is an easier frame than a cluttered photograph. At the 3B the registered HOST-BLIND reading is falsified by its own text arm — on the same rendered frames the caution text drives stop at 0.87 and the social text greets at 0.54, so perception and language conditioning transfer perfectly — and what fails to transfer is the *packet*: at s42 the packet equals

the habit on every mode, while the same doors lean 0.56–0.67 on photographs. The failure is ordered by door strength, the study’s recurring axis: s42’s weak doors die (0.00), s1042 sits at the threshold (0.23), and s2042’s near-perfect caution door travels at full strength (1.00). The prediction that transfer would track the host’s perception was wrong in the informative direction. The door’s injection is frame-blind, but its *effect* rides on frame-conditioned residual state, and rendered frames put the 3B’s stream where only a strong door’s direction still bites.

That turns the section into a prescription, and three registered cells wrote it (Table 10). Rung 3c doubled the steps: the strength lever that had bought hostile survival for free moves the domain gap and does not close it — s42 caution 0.00 $\rightarrow$ 0.31, s1042 0.23 $\rightarrow$ 0.18 — so the gap lives partly in the door’s *direction*, where in activation space its packet pushes, beyond how hard. Rung 3d put 80 rendered frames into the training mix at the registered step count: the mix closes the render gap outright (0.95/0.86) and pays for it in hostile robustness, s1042’s caution ownership under attack falling from 0.48 to 0.125 while calm conditioning survives. The bus had a domain budget, and the detail said what it was spent from: not the calm lean but the resistance margin. Rung 3e, the synthesis, trained the mixed doors at twice the steps, and every band passed with room — renders 0.86/1.00, calm packet shares retained or improved, and hostile ownership 0.85 against the registered door’s own 0.48. The budget was a *training* budget, not a bus-capacity budget: the same 256-dimensional packet at the same dose holds two domains and more margin than the single-domain door ever had. The lever algebra closes. Mix buys direction, steps buy margin, and they compound rather than compete; at the 3B the recommended door recipe is the mixed domain at twice the steps, and there it dominates the registered recipe on every axis measured.

Then the prescription was put to the other two hosts, and it did not survive as written (rung 3f, Table 11). At the 450M the recipe *trades*: it repairs the one seed whose registered doors did not travel (caution on renders 0.07 $\rightarrow$ 0.94) and breaks the two that did (1.00 $\rightarrow$ 0.36 and 0.44), halves one seed’s calm caution on photographs (1.00 $\rightarrow$ 0.50) and inverts the explore packet — DOMAIN LOST by the registered band. What was a training budget at the 3B is a capacity budget at the 450M: the same mix, steps, and bus cannot hold a second domain in a smaller host’s residual stream without dropping the first. At the 27B the registered doors already travel unaided (two of three seeds, caution leans 0.52/0.81/0.50, the packet again above the prompt on renders), and the recipe is mixed: it cures the suggestible s2042 door of §5 (hostile ownership 0.67 $\rightarrow$ 1.17, its grounding pulse 0.13 $\rightarrow$ 0.07, social awake on renders), doubles the middle seed’s margin (0.92 $\rightarrow$ 2.00), and damages seed 42, whose calm caution falls to 0.50 on photographs at the 27B exactly as at the 450M — the same seed, the same number, at both new hosts, with half of “stop for people” migrated into the pair’s constant door. The letter of the caution-lean band fails at the 27B for a reason the table makes visible: the mixed-domain constant door is no longer neutral on rendered person frames (it greets 0.85 at one seed and stops 0.94 at another where the registered constant doors proceed), so the lean’s denominator moves while the packet’s own stop share saturates (0.50/1.00/1.00). The honest reading is that the 27B’s mx2 packets travel at every seed and hold at two, and that mixing the domains moves *both* doors of the pair, which the 3B tolerated and the other hosts did not. The prescription therefore comes with a scale on it: at the onboard scale the recipe buys everything; at the creature scale keep single-domain doors and repair gap seeds one at a time; at the cortex scale the registered doors already travel and the recipe is a per-seed cure with a per-seed cost. Two cells are named and not registered: the seed-42 batch-order test, and a constant door trained on photographs only while the mode door sees the mix.

What the transfer story does not yet cover is stated as plainly as what it does. The rendered person is one identity in one static pose; rung 3 tests transfer mechanics, not detector generality, which COCO already established. Renders are not bit-reproducible across GPU or driver changes on the same machine, which the registrations from rung 3f onward handle by pinning scene identity

and the detector reading rather than pixel hashes. The gate of §7 has not been run on rendered episode streams or with a moving camera; the 27B has been put on renders but not yet through the gate; and the three-word action space keeps every result here at the level of behavioural modes, which is what the bus was built to carry and all it has been shown to carry.

figure	level	person score	accuracy	what it sees
capsule	l0	0.000	0.500	–
capsule	l1	0.000	0.500	–
mesh	mesh	0.908	0.995	–

Table 8: YOLO26s on capsule and photoscan renders. Person score is the mean person probability on present frames; descriptions are recorded detector outputs when available.

host	recipe	mode	lean seeds	pooled	grounding
450M	registered	social	0.50/0.41/0.49	0.47	0.00
450M	registered	caution	1.00/0.07/1.00	0.69	0.00
450M	registered	explore	0.00/0.00/-0.50	-0.17	–
3B	registered	social	0.02/0.08/0.08	0.06	0.00
3B	registered	caution	0.00/0.23/1.00	0.41	0.00
3B	registered	explore	0.00/0.00/0.00	0.00	–

Table 9: Unchanged registered doors on photoscan renders, seed order 42/1042/2042.

recipe	render lean	calm packet	hostile ownership
registered	0.00/0.23/1.00	0.27/1.00/1.00	0.00/0.48/0.83
x2	0.31/0.18	1.00	1.47
mx	0.95/0.86	0.97/0.90	0.35/0.12
mx2	0.86/1.00	0.90/1.00	0.29/0.85

Table 10: The 3B caution domain budget by door recipe. Registered has seeds 42/1042/2042; x2, mx, and mx2 have seeds 42/1042. Capsule host-probe discrimination is 0.267.

host	recipe	render packet	render habit	render lean	calm packet	hostile ownership
450M	registered	1.00/0.07/1.00	0.00/0.00/0.00	1.00/0.07/1.00	1.00/1.00/1.00	0.97/1.33/1.00
450M	mx2	0.50/1.00/1.00	0.14/0.06/0.56	0.36/0.94/0.44	0.50/0.97/1.00	-5.00/1.12/1.18
3B	registered	0.00/0.23/1.00	0.00/0.00/0.00	0.00/0.23/1.00	0.27/1.00/1.00	0.00/0.48/0.83
3B	mx2	1.00/1.00	0.14/0.00	0.86/1.00	0.90/1.00	0.29/0.85
27B	registered	0.55/0.82/0.50	0.03/0.01/0.00	0.52/0.81/0.50	1.00/1.00/1.00	1.17/0.92/0.67
27B	mx2	0.50/1.00/1.00	0.07/0.03/0.94	0.43/0.97/0.06	0.50/1.00/0.97	-2.00/1.17

Table 11: Rung 3f: the mixed-domain 2x-step recipe against the registered doors at every host, caution mode, seed order 42/1042/2042 (3B mx2: 42/1042). Render columns are stop shares and lean on photoscan-render person frames; calm packet is the stop share on COCO person frames; ownership is against the hostile prompt at dose 1x.

10 Numbers appendix

Every number in this scaffold is loaded from the JSON artifacts bundled under `note18/artifacts/`. Seed columns read the per-seed files; pooled columns read the no-suffix files. Registrations, admissibility bands, detector rules, checkpoint hashes, frame-manifest hashes, and run metadata remain in those artifacts. The source registrations and dated readings are in `tracks/vision-mode/PLAN.md`; this renderer does not copy claims out of prose or type numerical results into the manuscript.

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